

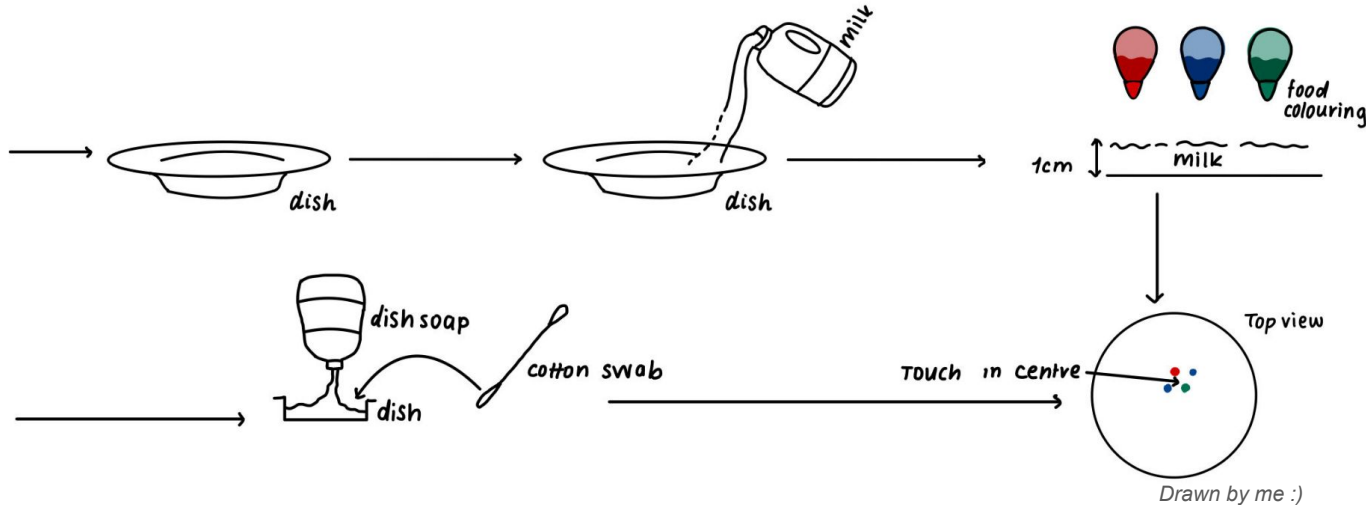
Milk Art Experiment

Equipment

- Whole milk.
- Food colouring (various colours).
- Dishwashing soap.
- Cotton swabs or Q-tips.
- Dish or shallow container.

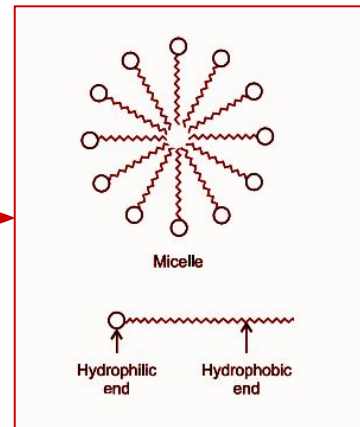
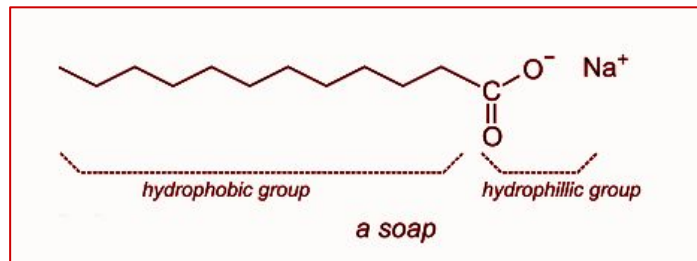
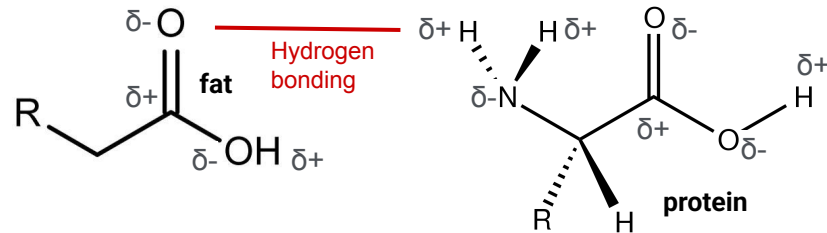
Method

1. Pour a thin layer of whole milk into the dish, covering the bottom evenly.
2. Add drops of different coloured food colouring to the surface of the milk. Space the drops apart from each other.
3. Dip the cotton swab or Q-tip into the dishwashing soap.
4. Gently touch the soapy cotton swab to the centre of one of the food colouring drops without stirring the milk.
5. Observe and describe the changes that occur on the surface of the milk.



Explanation

- Surface tension is a **cohesive** force.
- It causes the molecules at the **surface** of the liquid to stick together.
- **Fats** and **proteins** in milk contribute to its high surface tension.
- This surface tension **resists** the **penetration** of foreign substances.
- The introduction of **soap** disrupts the surface tension of milk.
- **Soap molecules** have unique properties, one end is **hydrophilic** and the other end is **hydrophobic**.
- When the soapy cotton swab is gently touched to the milk's surface, the **soap molecules start interacting with the milk components**.
- The **hydrophobic** end of the soap molecules is **attracted** to the **fat** molecules in the milk.
- Fats are composed of **hydrophobic** ends, forming structures known as **micelles**.



- This interaction causes the **fat** molecules to disperse into smaller droplets, **breaking down** the **cohesive layer** of fat at the surface.
- **Micelles** are aggregates of **soap** molecules where the **hydrophilic** ends face **outward**, interacting with **water molecules**.
- The **hydrophobic ends** cluster around the fat molecules.
 - *This is crucial in **emulsification**, where fats are dispersed in a water-based solution.*
- The food colouring, being **water-soluble**, is **carried along** with the movement of the dispersed fat droplets.
- The soap molecules facilitate the **suspension** of the food colouring in the milk.
- This **prevents** it from **sinking** or **settling**.
- The movement of soap through the milk **induces** a continuous **rearrangement** of fat and water components.
- This rearrangement combined with the suspended food colouring, leads to the creation of intricate colorful patterns on the milk's surface.